

BTC Market Intelligence Engine

Research & Architecture Brief — v6

Scope: BTCUSDT perpetual MVP

Status: Architecture baseline

Primary architectural principle: Hexagonal Architecture (Ports & Adapters)

1. Vision

Build a deterministic market-intelligence and research system for BTCUSDT perpetual that first maps the market, learns conditional market behavior, and only afterwards discovers, validates and selects strategies.

The system is not a “super strategy”. It is a measurable research machine that can discover which combinations of market state, regime, location and trigger contain repeatable edge.

2. Core principle

Market State → Regime → Conditional Behavior → Bias → Location → Trigger → Execution → Outcome → Research Feedback

The system must distinguish:

- **Bias:** what the broader evidence suggests.
- **Location:** where the market is relative to meaningful levels.
- **Trigger:** what observable event authorizes execution.
- **Execution:** entry, invalidation, target and management.
- **Outcome:** what actually happened.
- **Research:** whether the hypothesis and its components add measurable edge.

No trading concept is accepted merely because it is intuitively plausible.

3. Explicit architectural principle — Hexagonal Architecture

The implementation should follow the principles of **Hexagonal Architecture / Ports & Adapters**.

Order Flow / Aggression is a domain-facing capability fed by exchange-native trade and depth adapters. Bookmap is explicitly not a runtime dependency.

The central MIE domain must not depend directly on Binance, TradingView, PostgreSQL, DuckDB, Telegram, LLM providers, exchange SDKs or other infrastructure.

Domain core

The domain core contains the logic that should remain stable when infrastructure changes:

- Market State
- Regime
- Level / Location
- Bias
- Trigger
- Conditional Market Behavior
- Strategy hypotheses
- Strategy Suitability

- Risk
- Validation rules
- Edge / degradation evaluation

Application layer

Application services orchestrate use cases without owning external technology details:

- analyze current market
- replay historical market state
- run research experiment
- run backtest
- run walk-forward validation
- evaluate strategy suitability
- record research results
- generate an observability report

Inbound ports

Examples:

- `AnalyzeMarket`
- `ReplayMarket`
- `RunResearchExperiment`
- `EvaluateStrategy`
- `ValidateCandidate`
- `GenerateMarketReport`

Outbound ports

Examples:

- `MarketDataProvider`
- `HistoricalDataProvider`
- `FeatureStore`
- `StrategyRepository`
- `BacktestEngine`
- `ExecutionGateway`
- `AlertGateway`
- `ResearchResultStore`

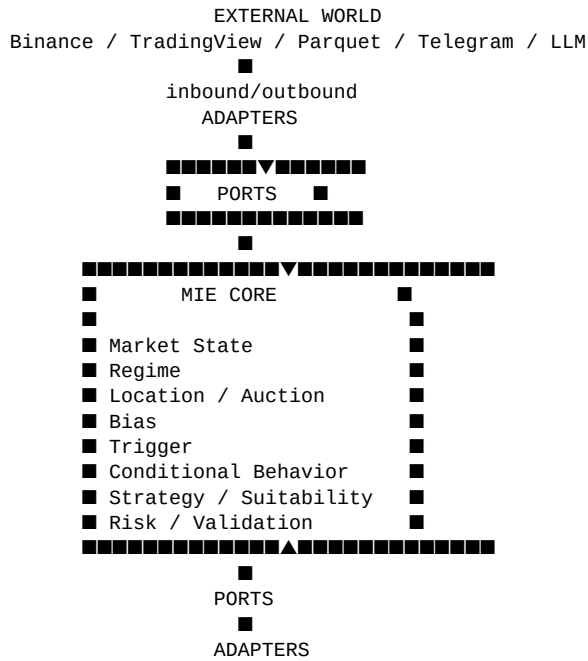
Adapters

Initial adapters may include:

- Binance market-data adapter
- Binance historical/replay adapter
- Parquet adapter
- DuckDB adapter
- TradingView/reference-definition adapter
- Telegram adapter
- LLM/agent adapter
- backtest-engine adapter

The core must be runnable without any single external adapter. This enables live processing, historical replay, backtesting and forward testing to use the same domain logic.

4. Target architecture



The architecture is deliberately dependency-inverted: infrastructure depends on domain-defined ports, not the other way around.

5. Live/replay path

```

Binance → ingestion adapter → MarketDataPort
        → deterministic Market State
        → Regime
        → Location / Bias
        → Trigger monitoring
        → Strategy Suitability
        → Observability / Egress
```

Historical replay uses the same domain processing path:

```

Parquet → HistoricalDataAdapter → same MarketDataPort
        → same Market State / Regime / Strategy logic
```

This is a critical design requirement: live and historical processing must not contain separate implementations of market logic.

6. Research path

```

Knowledge
↓
Hypothesis
↓
Experiment
↓
Replay / Backtest
↓
Feature contribution
↓
Refutation
↓
Validation
↓
Walk-forward
↓
Unseen forward test
↓
```

Strategy Registry



Live feedback

Agents/LLMs may propose hypotheses, plan experiments, synthesize results and challenge assumptions. Deterministic data, replay and validation remain authoritative.

7. Data Plane

Scope: BTCUSDT perpetual, one primary consumer.

Initial path:

Binance → Rust ingestion → immutable raw Parquet → DuckDB replay/research

Raw candidates:

- trades
- depth/order-book events
- klines/OHLCV
- open interest
- funding
- liquidations

Raw data is the source of truth. Derived features must be reproducible from raw data.

8. Market State

Canonical deterministic representation of current conditions.

Initial features:

- price, returns, velocity and range
- ATR(14)
- ATR percentile(200)
- OI, Δ OI and OI velocity
- trade delta and CVD
- bid/ask volume
- order-book imbalance
- liquidity added/removed and concentration
- absorption/exhaustion candidates
- volume profile: POC, VAH, VAL, HVN, LVN
- structural highs/lows
- SFP / sweep detection
- CVD/OI/price divergences
- effort-vs-result relationships
- multi-timeframe context

9. Regime

Initial ATR percentile regime:

- **LOW:** 0–25
- **MEDIUM:** 26–50
- **HIGH:** 51–75
- **EXTREME:** 76–100

The raw percentile is retained in addition to the label.

ATR percentile is contextual information, not a direct entry signal.

Future regime dimensions must be validated empirically: trend/range/transition, compression/expansion, directional efficiency, liquidity environment, OI, funding, session/time-of-day, location and regime duration.

10. Location and auction state

Maintain and score:

- POC
- VAH / VAL
- HVN / LVN
- structural highs/lows
- liquidity clusters
- prior sweeps
- SFP/rejection zones
- VWAP and other validated levels

For range trading, explicitly distinguish:

- inside value
- at value edge
- outside value
- breakout
- failed breakout / failed auction
- failed reclaim
- acceptance

The system continues to monitor a location even when no trade is authorized.

11. Bias / Location / Trigger separation

A directional bias does not authorize execution.

Example:

Bias:
Range + price near VAH + HIGH volatility

Location:
VAH / upper value edge

Trigger candidates:

- failed auction
- SFP
- absorption
- aggression with no price progress
- CVD divergence
- OI expansion followed by rejection
- breakout + acceptance

The same bias and location can therefore be tested with multiple trigger families.

12. Conditional Market Behavior

Before strategy optimization, measure what tends to happen after a given Market State + Regime + Location + Trigger context.

Outputs can include:

- upside/downside threshold first
- MFE / MAE
- time-to-event
- level revisit probability
- invalidation probability
- excursion distribution
- outcome conditional on volatility regime
- outcome conditional on location and trigger

This layer is not a BUY/SELL classifier.

13. Strategy Research Lab

The lab continuously:

1. generates hypotheses
2. formalizes them
3. runs deterministic replay/backtests
4. evaluates feature contribution
5. performs counter-hypothesis/refutation tests
6. evaluates confirmation cost
7. validates survivors
8. runs walk-forward tests
9. runs unseen forward tests
10. registers validated candidates
11. monitors live degradation

No feature is assumed useful merely because it sounds plausible.

14. Strategy Registry & Suitability

Lifecycle:

RESEARCH → BACKTESTED → WALK_FORWARD → PAPER → VALIDATED → ACTIVE →
DEGRADING/SUSPENDED → DEPRECATED

At runtime:

```
Current Market State
+
Current Regime
+
Current Location
+
Available validated triggers
↓
Strategy Suitability
```

The system may legitimately return:

NO SUITABLE STRATEGY

15. Edge decay and risk

A validated strategy can stop working.

Track:

- rolling expectancy
- profit factor
- win rate
- MFE/MAE
- drawdown
- performance by regime
- performance by location
- performance by trigger
- deviation from validation distributions

Risk is a separate subsystem and must be testable.

16. LLM / Agent boundary

LLMs are research operators and adapters, not numerical authorities.

Allowed:

- hypothesis generation
- research planning
- result synthesis
- challenge/refutation
- experiment prioritization

Not authoritative for:

- raw market facts
- deterministic feature values
- backtest metrics
- validation decisions

17. Egress

First practical egress: Telegram.

Example:

```
Alert: BTC entered a monitored VAH zone.  
Guide: Failed-auction short is only confirmed if the defined  
acceptance/rejection conditions occur.  
Invalidation: explicit level/condition.
```

Autonomous execution remains outside the MVP.

18. Two feedback loops

Market loop

Binance → Market State → Regime → Observation → Outcome → Knowledge

Research loop

Knowledge → Hypothesis → Experiment → Validation → Registry → Suitability → Live feedback

19. First milestone

Binance → Rust → raw Parquet → deterministic Market State → ATR/OI/CVD/VP/structure → Regime → conditional outcome dataset → DuckDB replay/research → Strategy Research Lab → Telegram

The first implementation should validate the architecture, not prematurely optimize a trading strategy.

20. Explicitly deferred

- multi-asset scale
- ClickHouse
- autonomous execution
- one fixed strategy
- premature parameter optimization
- LLM-driven numerical truth
- unvalidated discretionary concepts
- strategy selection based only on backtest performance